

JAVASCRIPT

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JavaScript is used in millions of Web pages to add functionality, validate forms, detect browsers, and much more.

Introduction to JavaScript

JavaScript is used in millions of Web pages to improve the design, validate forms, detect browsers, create cookies, and much more.

JavaScript is the most popular scripting language on the Internet, and works in all major browsers, such as Internet Explorer, Mozilla Firefox, and Opera.

What is JavaScript?

- JavaScript was designed to add interactivity to HTML pages
- JavaScript is a scripting language
- A scripting language is a lightweight programming language
- JavaScript is usually embedded directly into HTML pages
- JavaScript is an interpreted language (means that scripts execute without preliminary compilation)
- Everyone can use JavaScript without purchasing a license

Java and JavaScript are two completely different languages in both concept and design!

Java (developed by Sun Microsystems) is a powerful and much more complex programming language - in the same category as C and C++.

What can a JavaScript Do ?

- **JavaScript gives HTML designers a programming tool** - HTML authors are normally not programmers, but JavaScript is a scripting language with a very simple syntax! Almost anyone can put small "snippets" of code into their HTML pages
- **JavaScript can put dynamic text into an HTML page** - A JavaScript statement like this: `document.write("<h1>" + name + "</h1>")` can write a variable text into an HTML page
- **JavaScript can react to events** - A JavaScript can be set to execute when something happens, like when a page has finished loading or when a user clicks on an HTML element
- **JavaScript can read and write HTML elements** - A JavaScript can read and change the content of an HTML element
- **JavaScript can be used to validate data** - A JavaScript can be used to validate form data before it is submitted to a server. This saves the server from extra processing
- **JavaScript can be used to detect the visitor's browser** - A JavaScript can be used to detect the visitor's browser, and - depending on the browser - load another page specifically designed for that browser

- **JavaScript can be used to create cookies** - A JavaScript can be used to store and retrieve information on the visitor's computer.

JavaScript Variables

Variables are "containers" for storing information.

JavaScript variables are used to hold values or expressions.

A variable can have a short name, like x, or a more descriptive name, like carname.

Rules for JavaScript variable names:

- Variable names are case sensitive (y and Y are two different variables)
- Variable names must begin with a letter or the underscore character

Note: Because JavaScript is case-sensitive, variable names are case-sensitive.

Example

A variable's value can change during the execution of a script. You can refer to a variable by its name to display or change its value.

```
<html>
<body>
<script type="text/javascript">
var firstname;
firstname="Welcome";
document.write(firstname);
document.write("<br />");
firstname="XYZ";
document.write(firstname);
</script>
```

<p>The script above declares a variable, assigns a value to it, displays the value, change the value, and displays the value again.</p>

```
</body>
</html>
```

Output :

Welcome
XYZ

The script above declares a variable, assigns a value to it, displays the value, change the value, and displays the value again.

Declaring (Creating) JavaScript Variables

Creating variables in JavaScript is most often referred to as "declaring" variables.

You can declare JavaScript variables with the **var statement**:

```
var x;  
var carname;
```

After the declaration shown above, the variables are empty (they have no values yet).

However, you can also assign values to the variables when you declare them:

```
var x=5;  
var carname="Scorpio";
```

After the execution of the statements above, the variable **x** will hold the value **5**, and **carname** will hold the value **Scorpio**.

Note: When you assign a text value to a variable, use quotes around the value.

Assigning Values to Undeclared JavaScript Variables

If you assign values to variables that have not yet been declared, the variables will automatically be declared.

These statements:

```
x=5;  
carname="Scorpio";
```

have the same effect as:

```
var x=5;  
var carname="Scorpio";
```

Redeclaring JavaScript Variables

If you redeclare a JavaScript variable, it will not lose its original value.

```
var x=5;  
var x;
```

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After the execution of the statements above, the variable x will still have the value of 5. The value of x is not reset (or cleared) when you redeclare it.

DataTypes

- **Numbers** - are values that can be processed and calculated. You don't enclose them in quotation marks. The numbers can be either positive or negative.
- **Strings** - are a series of letters and numbers enclosed in quotation marks. JavaScript uses the string literally; it doesn't process it. You'll use strings for text you want displayed or values you want passed along.
- **Boolean (true/false)** - lets you evaluate whether a condition meets or does not meet specified criteria.
- **Null** - is an empty value. **null** is not the same as 0 -- 0 is a real, calculable number, whereas **null** is the **absence of any value**.

Data Types

TYPE	EXAMPLE
Numbers	Any number, such as 17, 21, or 54e7
Strings	"Greetings!" or "Fun"
Boolean	Either true or false
Null	A special keyword for exactly that – the null value (that is, nothing)

JavaScript Arithmetic

As with algebra, you can do arithmetic operations with JavaScript variables:

```
y=x-5;  
z=y+5;
```

JavaScript Operators

The operator = is used to assign values.

The operator + is used to add values.

The assignment operator = is used to assign values to JavaScript variables.

The arithmetic operator + is used to add values together.

```
y=5;  
z=2;  
x=y+z;
```

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The value of x, after the execution of the statements above is 7.

JavaScript Arithmetic Operators

Arithmetic operators are used to perform arithmetic between variables and/or values.

Given that **y=5**, the table below explains the arithmetic operators:

Operator	Description	Example	Result
+	Addition	x=y+2	x=7
-	Subtraction	x=y-2	x=3
*	Multiplication	x=y*2	x=10
/	Division	x=y/2	x=2.5
%	Modulus (division remainder)	x=y%2	x=1
++	Increment	x=++y	x=6
--	Decrement	x=--y	x=4

JavaScript Assignment Operators

Assignment operators are used to assign values to JavaScript variables.

Given that **x=10** and **y=5**, the table below explains the assignment operators:

Operator	Example	Same As	Result
=	x=y		x=5
+=	x+=y	x=x+y	x=15
-=	x-=y	x=x-y	x=5
=	x=y	x=x*y	x=50
/=	x/=y	x=x/y	x=2
%=	x%=y	x=x%y	x=0

The + Operator Used on Strings

The + operator can also be used to add string variables or text values together.

To add two or more string variables together, use the + operator.

```
txt1="What a very";  
txt2="nice day";  
txt3=txt1+txt2;
```

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After the execution of the statements above, the variable txt3 contains "What a verynice day".

To add a space between the two strings, insert a space into one of the strings:

```
txt1="What a very ";  
txt2="nice day";  
txt3=txt1+txt2;
```

or insert a space into the expression:

```
txt1="What a very";  
txt2="nice day";  
txt3=txt1+" "+txt2;
```

After the execution of the statements above, the variable txt3 contains:

"What a very nice day"

Adding Strings and Numbers

Look at these examples:

```
x=5+5;  
document.write(x);  
  
x="5"+"5";  
document.write(x);  
  
x=5+"5";  
document.write(x);  
  
x="5"+5;  
document.write(x);
```

The rule is:

If you add a number and a string, the result will be a string.

JavaScript Comparison and Logical Operators

Comparison and Logical operators are used to test for true or false.

Comparison Operators

Comparison operators are used in logical statements to determine equality or difference between variables or values.

Given that **x=5**, the table below explains the comparison operators:

Operator	Description	Example
==	is equal to	x==8 is false
===	is exactly equal to (value and type)	x===5 is true x=== "5" is false
!=	is not equal	x!=8 is true
>	is greater than	x>8 is false
<	is less than	x<8 is true
>=	is greater than or equal to	x>=8 is false
<=	is less than or equal to	x<=8 is true

How Can it be Used

Comparison operators can be used in conditional statements to compare values and take action depending on the result:

```
if (age<18) document.write("Too young");
```

You will learn more about the use of conditional statements in the next chapter of this tutorial.

Logical Operators

Logical operators are used to determine the logic between variables or values.

Given that **x=6 and y=3**, the table below explains the logical operators:

Operator	Description	Example
&&	and	(x < 10 && y > 1) is true
	or	(x==5 y==5) is false
!	not	!(x==y) is true

Conditional Operator

JavaScript also contains a conditional operator that assigns a value to a variable based on some condition.

Syntax

```
variablename=(condition)?value1:value2
```

Example

```
greeting=(visitor=="PRES")?"Dear President ":"Dear ";
```

If the variable **visitor** has the value of "PRES", then the variable **greeting** will be assigned the value "Dear President " else it will be assigned "Dear".

Conditional Statements

Very often when you write code, you want to perform different actions for different decisions. You can use conditional statements in your code to do this.

In JavaScript we have the following conditional statements:

- **if statement** - use this statement if you want to execute some code only if a specified condition is true
- **if...else statement** - use this statement if you want to execute some code if the condition is true and another code if the condition is false
- **if...else if....else statement** - use this statement if you want to select one of many blocks of code to be executed
- **switch statement** - use this statement if you want to select one of many blocks of code to be executed

If Statement

You should use the if statement if you want to execute some code only if a specified condition is true.

Syntax

```
if (condition)
{
  code to be executed if condition is true
}
```

Note that if is written in lowercase letters. Using uppercase letters (IF) will generate a JavaScript error!

Example 1

```
<script type="text/javascript">
//Write a "Good morning" greeting if
//the time is less than 10
var d=new Date();
var time=d.getHours();
```

```
if (time<10)
{
document.write("<b>Good morning</b>");
}
</script>
```

Example 2

```
<script type="text/javascript">
//Write "Lunch-time!" if the time is 11
var d=new Date();
var time=d.getHours();

if (time==11)
{
document.write("<b>Lunch-time!</b>");
}
</script>
```

Note: When **comparing** variables you must always use two equals signs next to each other (==)!

Notice that there is no ..else.. in this syntax. You just tell the code to execute some code **only if the specified condition is true**.

If...else Statement

If you want to execute some code if a condition is true and another code if the condition is not true, use the if....else statement.

Syntax

```
if (condition)
{
code to be executed if condition is true
}
else
{
code to be executed if condition is not true
}
```

Example

```
<script type="text/javascript">
//If the time is less than 10,
//you will get a "Good morning" greeting.
//Otherwise you will get a "Good day" greeting.
var d = new Date();
```

```
var time = d.getHours();

if (time < 10)
{
document.write("Good morning!");
}
else
{
document.write("Good day!");
}
</script>
```

If...else if...else Statement

You should use the if....else if...else statement if you want to select one of many sets of lines to execute.

Syntax

```
if (condition1)
{
code to be executed if condition1 is true
}
else if (condition2)
{
code to be executed if condition2 is true
}
else
{
code to be executed if condition1 and
condition2 are not true
}
```

Example

```
<script type="text/javascript">
var d = new Date()
var time = d.getHours()
if (time<10)
{
document.write("<b>Good morning</b>");
}
else if (time>10 && time<16)
{
document.write("<b>Good day</b>");
}
else
```

```
{  
document.write("<b>Hello World!</b>");  
}  
</script>
```

The JavaScript Switch Statement

You should use the switch statement if you want to select one of many blocks of code to be executed.

Syntax

```
switch(n)  
{  
case 1:  
    execute code block 1  
    break;  
case 2:  
    execute code block 2  
    break;  
default:  
    code to be executed if n is  
    different from case 1 and 2  
}
```

This is how it works: First we have a single expression n (most often a variable), that is evaluated once. The value of the expression is then compared with the values for each case in the structure. If there is a match, the block of code associated with that case is executed. Use **break** to prevent the code from running into the next case automatically.

Example

```
<script type="text/javascript">  
//You will receive a different greeting based  
//on what day it is. Note that Sunday=0,  
//Monday=1, Tuesday=2, etc.  
var d=new Date();  
theDay=d.getDay();  
switch (theDay)  
{  
case 5:  
    document.write("Finally Friday");  
    break;  
case 6:  
    document.write("Super Saturday");  
    break;  
case 0:  
    document.write("Sleepy Sunday");
```

```
break;
default:
  document.write("I'm looking forward to this weekend!");
}
</script>
```

JavaScript Controlling(Looping) Statements

Loops in JavaScript are used to execute the same block of code a specified number of times or while a specified condition is true.

JavaScript Loops

Very often when you write code, you want the same block of code to run over and over again in a row. Instead of adding several almost equal lines in a script we can use loops to perform a task like this.

In JavaScript there are two different kind of loops:

- **for** - loops through a block of code a specified number of times
- **while** - loops through a block of code while a specified condition is true

The for Loop

The for loop is used when you know in advance how many times the script should run.

Syntax

```
for (var=startvalue;var<=endvalue;var=var+increment)
{
  code to be executed
}
```

Example

Explanation: The example below defines a loop that starts with `i=0`. The loop will continue to run as long as `i` is less than, or equal to 10. `i` will increase by 1 each time the loop runs.

Note: The increment parameter could also be negative, and the `<=` could be any comparing statement.

```
<html>
<body>
<script type="text/javascript">
var i=0;
for (i=0;i<=10;i++)
{
```

```
document.write("The number is " + i);  
document.write("<br />");  
}  
</script>  
</body>  
</html>
```

Result

```
The number is 0  
The number is 1  
The number is 2  
The number is 3  
The number is 4  
The number is 5  
The number is 6  
The number is 7  
The number is 8  
The number is 9  
The number is 10
```

JavaScript While Loop

Loops in JavaScript are used to execute the same block of code a specified number of times or while a specified condition is true.

The while loop

The while loop is used when you want the loop to execute and continue executing while the specified condition is true.

```
while (var<=endvalue)  
{  
    code to be executed  
}
```

Note: The <= could be any comparing statement.

Example

Explanation: The example below defines a loop that starts with i=0. The loop will continue to run as long as **i** is less than, or equal to 10. **i** will increase by 1 each time the loop runs.

```
<html>
```

```
<body>
<script type="text/javascript">
var i=0;
while (i<=10)
{
document.write("The number is " + i);
document.write("<br />");
i=i+1;
}
</script>
</body>
</html>
```

Result

```
The number is 0
The number is 1
The number is 2
The number is 3
The number is 4
The number is 5
The number is 6
The number is 7
The number is 8
The number is 9
The number is 10
```

The do...while Loop

The do...while loop is a variant of the while loop. This loop will always execute a block of code ONCE, and then it will repeat the loop as long as the specified condition is true. This loop will always be executed at least once, even if the condition is false, because the code is executed before the condition is tested.

```
do
{
    code to be executed
}
while (var<=endvalue);
```

Example

```
<html>
<body>
<script type="text/javascript">
```

```
var i=0;
do
{
document.write("The number is " + i);
document.write("<br />");
i=i+1;
}
while (i<0);
</script>
</body>
</html>
```

Result

The number is 0

JavaScript Break and Continue

There are two special statements that can be used inside loops: break and continue.

JavaScript break and continue Statements

There are two special statements that can be used inside loops: break and continue.

Break

The break command will break the loop and continue executing the code that follows after the loop (if any).

Example

```
<html>
<body>
<script type="text/javascript">
var i=0;
for (i=0;i<=10;i++)
{
if (i==3)
{
break;
}
document.write("The number is " + i);
document.write("<br />");
}
</script>
</body>
```



```
</html>
```

Result

```
The number is 0  
The number is 1  
The number is 2
```

Continue

The continue command will break the current loop and continue with the next value.

Example

```
<html>  
<body>  
<script type="text/javascript">  
var i=0  
for (i=0;i<=10;i++)  
{  
if (i==3)  
{  
continue;  
}  
document.write("The number is " + i);  
document.write("<br />");  
}  
</script>  
</body>  
</html>
```

Result

```
The number is 0  
The number is 1  
The number is 2  
The number is 4  
The number is 5  
The number is 6  
The number is 7  
The number is 8  
The number is 9  
The number is 10
```

JavaScript Functions

A function (also known as a *method*) is a self-contained piece of code that performs a particular "function". You can recognise a function by its format - it's a piece of descriptive text, followed by open and close brackets. A function is a reusable code-block that will be executed by an event, or when the function is called.

To keep the browser from executing a script when the page loads, you can put your script into a function.

A function contains code that will be executed by an event or by a call to that function.

You may call a function from anywhere within the page (or even from other pages if the function is embedded in an external .js file).

Functions can be defined both in the <head> and in the <body> section of a document. However, to assure that the function is read/loaded by the browser before it is called, it could be wise to put it in the <head> section.

Example

```
<html>
<head>
<script type="text/javascript">
function displaymessage()
{
alert("Hello World!");
}
</script>
</head>
<body>
<form>
<input type="button" value="Click me!"
onclick="displaymessage()" >
</form>
</body>
</html>
```

If the line: `alert("Hello world!!")` in the example above had not been put within a function, it would have been executed as soon as the line was loaded. Now, the script is not executed before the user hits the button. We have added an `onClick` event to the button that will execute the function `displaymessage()` when the button is clicked.

You will learn more about JavaScript events in the JS Events chapter.

How to Define a Function

The syntax for creating a function is:

```
function functionname(var1,var2,...,varX)
{
some code
}
```

var1, var2, etc are variables or values passed into the function. The { and the } defines the start and end of the function.

Note: A function with no parameters must include the parentheses () after the function name:

```
function functionname()
{
some code
}
```

Note: Do not forget about the importance of capitals in JavaScript! The word function must be written in lowercase letters, otherwise a JavaScript error occurs! Also note that you must call a function with the exact same capitals as in the function name.

The return Statement

The return statement is used to specify the value that is returned from the function.

So, functions that are going to return a value must use the return statement.

Example

The function below should return the product of two numbers (a and b):

```
function prod(a,b)
{
  x=a*b;
  return x;
}
```

When you call the function above, you must pass along two parameters:

```
product=prod(2,3);
```

The returned value from the `prod()` function is 6, and it will be stored in the variable called `product`.

The Lifetime of JavaScript Variables

When you declare a variable within a function, the variable can only be accessed within that function. When you exit the function, the variable is destroyed. These variables are called local variables. You can have local variables with the same name in different functions, because each is recognized only by the function in which it is declared.

If you declare a variable outside a function, all the functions on your page can access it. The lifetime of these variables starts when they are declared, and ends when the page is closed.

What is an Event?

Event Handlers

Event Handlers are JavaScript methods, i.e. functions of objects, that allow us as JavaScript programmers to control what happens when events occur.

Directly or indirectly, an **Event** is always the result of something a user does. For example, we've already seen Event Handlers like **onClick** and **onMouseOver** that respond to mouse actions. Another type of Event, an internal change-of-state to the page (**completion** of loading or **leaving** the page). An **onLoad** Event can be considered an indirect result of a user action.

Although we often refer to Events and Event Handlers interchangeably, it's important to keep in mind the distinction between them. An **Event** is merely something that happens - something that it is initiated by an **Event Handler** (**onClick**, **onMouseOver**, etc...).

The elements on a page which can trigger events are known as "**targets**" or "**target elements**," and we can easily understand how a button which triggers a *Click* event is a target element for this event. Typically, events are defined through the use of Event Handlers, which are bits of script that tell the browser what to do when a particular event occurs at a particular target. These Event Handlers are commonly written as attributes of the target element's HTML tag.

The Event Handler for a *Click* event at a form field button element is quite simple to understand:

```
<INPUT TYPE="button" NAME="click1" VALUE="Click me for fun!"  
onClick="event_handler_code">
```

The `event_handler_code` portion of this example is any valid JavaScript and it will be executed when the specified event is triggered at this target element. This particular topic will be continued in [Incorporating JavaScripts into your HTML pages](#).

There are "three different ways" that Event Handlers can be used to trigger Events or Functions.

Method 1 (Link Events):

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Places an Event Handler as an attribute within an `<A HREF= >` tag, like this:

```
<A HREF="foo.html" onMouseOver="doSomething()"> ... </A>
```

You can use an Event Handler located within an `<A HREF= >` tag to make either an image or a text link respond to a mouseover Event. Just enclose the image or text string between the `<A HREF= >` and the `</A>` tags.

Whenever a user clicks on a link, or moves her cursor over one, JavaScript is sent a **Link Event**. One Link Event is called **onClick**, and it gets sent whenever someone clicks on a link. Another link event is called **onMouseOver**. This one gets sent when someone moves the cursor over the link.

You can use these events to affect what the user sees on a page. Here's an example of [how to use link events](#). Try it out, View Source, and we'll go over it.

```
<A HREF="javascript:void()"
  onClick="open('index.htm', 'links', 'height=200,width=200');">How to Use Link Events
</A>
```

The first interesting thing is that there are no `<SCRIPT>` tags. That's because anything that appears in the quotes of an **onClick** or an **onMouseOver** is automatically interpreted as JavaScript. In fact, because semicolons mark the end of statements allowing you to write entire JavaScripts in one line, you can fit an entire JavaScript program between the quotes of an **onClick**. It'd be ugly, but you could do it.

Here are the three lines of interest:

1. `<A HREF="#" onClick="alert('Ooo, do it again!');">Click on me!</A>`
2. `<A HREF="javascript:void()" onClick="alert('Ooo, do it again!');">Click on me!</A>`
3. `<A HREF="javascript:alert('Ooo, do it again!');" >Click on me!</A>`

In the first example we have a normal `<A>` tag, but it has the magic `onClick=""` element, which says, "When someone clicks on this link, run the little bit of JavaScript between my quotes." Notice, there's even a terminating semicolon at the end of the alert. **Question:** is this required? NO.

Let's go over each line:

1. `HREF="#"` tells the browser to look for the anchor `#`, but there is no anchor `"#"`, so the browser reloads the page and goes to top of the page since it couldn't find the anchor.
2. `<A HREF="javascript:void()"` tells the browser not to go anywhere - it "deadens" the link when you click on it. `HREF="javascript:` is the way to call a function when a link (hyperlink or an HREFed image) is clicked.
3. `HREF="javascript:alert('Ooo, do it again!')"` here we kill two birds with one stone. The default behavior of a hyperlink is to click on it. By clicking on the link we call the window Method `alert()` and also at the same time "deadens" the link.

The next line is

```
<A HREF="javascript:void()" onMouseOver="alert('Hee hee!');">  
Mouse over me!  
</A>
```

This is just like the first line, but it uses an **onMouseOver** instead of an **onClick**.

Method 2 (Actions within FORMs):

The second technique we've seen for triggering a Function in response to a mouse action is to place an **onClick** Event Handler inside a button type form element, like this:

```
<FORM>  
  <INPUT TYPE="button" onClick="doSomething()">  
</FORM>
```

While any JavaScript statement, methods, or functions can appear inside the quotation marks of an Event Handler, typically, the JavaScript script that makes up the Event Handler is actually a call to a function defined in the header of the document or a single JavaScript command. Essentially, though, anything that appears inside a command block (inside curly braces {}) can appear between the quotation marks.

For instance, if you have a form with a text field and want to call the function **checkField()** whenever the value of the text field changes, you can define your text field as follows:

```
<INPUT TYPE="text" onChange="checkField(this)">
```

Nonetheless, the entire code for the function could appear in quotation marks rather than a function call:

```
<INPUT TYPE="text" onChange="if (this.value <= 5) {  
  alert('Please enter a number greater than 5');  
}">
```

To separate multiple commands in an Event Handler, use semicolons

```
<INPUT TYPE="text" onChange="alert('Thanks for the entry.');" ;  
confirm('Do you want to continue?');">
```

The advantage of using functions as Event Handlers, however, is that you can use the same Event Handler code for multiple items in your document and, functions make your code easier to read and understand.

Method 3 (BODY onLoad & onUnload):

The third technique is to use an Event Handler to ensure that all required objects are defined involve the **onLoad** and **onUnload**. These Event Handlers are defined in the **<BODY>** or **<FRAMESET>** tag of an HTML file and are invoked when the document or frameset are fully loaded or unloaded. If you set a flag

within the **onLoad** Event Handler, other Event Handlers can test this flags to see if they can safely run, with the knowledge that the document is fully loaded and all objects are defined. For example:

```
<SCRIPT>

var loaded = false;

function doit() {
    // alert("Everything is \"loaded\" and loaded = " + loaded);
    alert("Everything is \"loaded\" and loaded = ' + loaded);
}

</SCRIPT>

<BODY onLoad="loaded = true;">
-- OR --
<BODY onLoad="window.loaded = true;">

<FORM>
    <INPUT TYPE="button" VALUE="Press Me"
        onClick="if (loaded == true) doit();">
-- OR --
    <INPUT TYPE="button" VALUE="Press Me"
        onClick="if (window.loaded == true) doit();">
-- OR --
    <INPUT TYPE="button" VALUE="Press Me"
        onClick="if (loaded) doit();">
</FORM>

</BODY>
```

The **onLoad** Event Handler is executed when the document or frameset is fully loaded, which means that all images have been downloaded and displayed, all subframes have loaded, any Java Applets and Plugins (Navigator) have started running, and so on. The **onUnload** Event Handler is executed just before the page is unloaded, which occurs when the browser is about to move on to a new page. Be aware that when you are working with multiple frames, there is no guarantee of the order in which the **onLoad** Event Handler is invoked for the various frames, except that the Event Handlers for the parent frame is invoked after the Event Handlers of all its children frames -- This will be discussed in detail in **Week 8**.

Setting the bgColor Property

The [first example](#) allows the user to change the color by clicking buttons, while the [second example](#) allows you to change colors by using drop down boxes.

Event Handlers

JAVASCRIPT Notes

EVENT	DESCRIPTION
onAbort	the user cancels loading of an image
onBlur	input focus is removed from a form element (when the user clicks outside the field) or focus is removed from a window
onClick	the user clicks on a link or form element
onChange	the value of a form field is changed by the user
onError	an error happens during loading of a document or image
onFocus	input focus is given to a form element or a window
onLoad	once a page is loaded, NOT while loading
onMouseOut	the user moves the pointer off of a link or clickable area of an image map
onMouseOver	the user moves the pointer over a hypertext link
onReset	the user clears a form using the Reset button
onSelect	the user selects a form element's field
onSubmit	a form is submitted (ie, when the users clicks on a submit button)
onUnload	the user leaves a page

Note: *Input focus* refers to the act of clicking on or in a form element or field. This can be done by clicking in a text field or by tabbing between text fields.

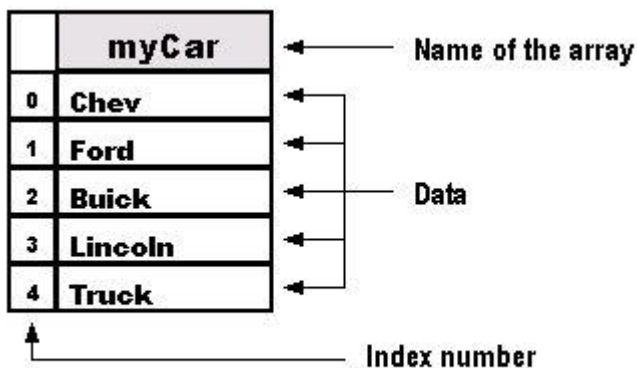
Which Event Handlers Can Be Used

OBJECT	EVENT HANDLERS AVAILABLE
Button element	onClick, onMouseOver
Checkbox	onClick
Clickable ImageMap area	onClick, onMouseOver, onMouseOut
Document	onLoad, onUnload, onError
Form	onSubmit, onReset
Framesets	onBlur, onFocus
Hypertext link	onClick, onMouseOver, onMouseOut
Image	onLoad, onError, onAbort

Radio button	onClick
Reset button	onClick
Selection list	onBlur, onChange, onFocus
Submit button	onClick
TextArea element	onBlur, onChange, onFocus, onSelect
Text element	onBlur, onChange, onFocus, onSelect
Window	onLoad, onUnload, onBlur, onFocus

JavaScript Arrays

An array object is used to create a database-like structure within a script. Grouping data points (*array elements*) together makes it easier to access and use the data in a script. There are methods of accessing actual databases (which are beyond the scope of this series) but here we're talking about small amounts of data.



Comparison of an array to a column of data

An array can be viewed like a column of data in a spreadsheet. The name of the array would be the same as the name of the column. Each piece of data (*element*) in the array is referred to by a number (*index*), just like a row number in a column.

An array is an *object*. Earlier, I said that an object is a thing, a collection of properties (*array elements*, in this case) grouped together.

You can name an array using the same format as a variable, a function or an object. Remember our basic rules: The first character cannot be a number, you cannot use a reserved word, and you cannot use spaces. Also, be sure to remember that the name of the array object is capitalized, e.g. **Array**.

The JavaScript interpreter uses numbers to access the collection of elements (i.e. the data) in an array. Each index number (as it is the number of the data in the array's index) refers to a specific piece of data in the array, similar to an ID number. It's important to remember that the index numbering of the data starts at "0." So, if you have 8 elements, the first element will be numbered "0" and the last one will be "7."

Elements can be of any type: character string, integer, Boolean, or even another array. An array can even have different types of elements within the same array. Each element in the

array is accessed by placing its index number in brackets, i.e. `myCar[4]`. This would mean that we are looking for data located in the array `myCar` which has an index of "4." Since the numbering of an index starts at "0," this would actually be the fifth index. For instance, in the following array,

```
var myCar = new Array("Chev","Ford","Buick","Lincoln","Truck");
alert(myCar[4])
```

the data point with an index of "4" would be `Truck`. In this example, the indexes are numbered as follows: 0=`Chev`, 1=`Ford`, 2=`Buick`, 3=`Lincoln`, and 4=`Truck`. When creating loops, it's much easier to refer to a number than to the actual data itself.

The Size of the Array

The size of an array is determined by either the actual number of elements it contains or by actually specifying a given size. You don't need to specify the size of the array. Sometimes, though, you may want to pre-set the size, e.g.:

```
var myCar = new Array(20);
```

That would pre-size the array with 20 elements. You might pre-size the array in order to set aside the space in memory.

Multidimensional Arrays

This type of an array is similar to [parallel arrays](#). In a multidimensional array, instead of creating two or more arrays in tandem as we did with the parallel array, we create an array with several levels or "dimensions." Remember [our example](#) of a spreadsheet with rows and columns? This time, however, we have a couple more columns.

	maker	model	color
0	Gibson	Les Paul	Sunburst
1	Fender	Stratocaster	Black
2	Martin	D-28	Mahogany
3	Takamine	EG330SC	Spruce

Comparison of a multidimensional array to a column of data

Multidimensional arrays can be created in different ways. Let's look at one of these methods. First, we create the main array, which is similar to what we did with previous arrays.

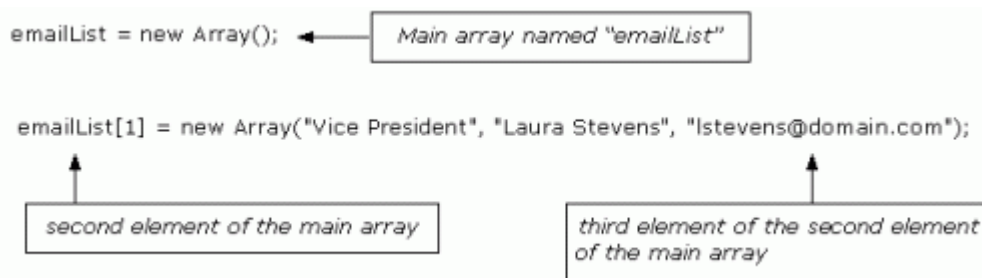
```
var emailList = new Array();
```

Next, we create arrays for elements of the main array:

```
emailList[0] = new Array("President", "Paul Smith", "psmith@domain.com");  
emailList[1] = new Array("Vice President", "Laura Stevens", "lstevens@domain.com");  
emailList[2] = new Array("General Manager", "Mary Larsen", "mlarsen@domain.com");  
emailList[3] = new Array("Sales Manager", "Bob Lark", "blark@domain.com");
```

In this script we created "sub arrays" or arrays from another level or "dimension." We used the name of the main array and gave it an index number (e.g., emailList[0]). Then we created a new instance of an array and gave it a value with three elements.

In order to access a single element, we need to use a double reference. For example, to get the e-mail address for the Vice President in our example above, access the third element "[2]" of the second element "[1]" of the array named emailList.



It would be written like this:

```
var vpEmail = emailList[1][2]  
alert("The address is: "+ vpEmail)
```

1. We declared a variable, named it emailList, and initialized it with a value of a new instance of an array.
2. Next, we created an array for each of the elements within the original array. Each of the new arrays contained three elements.
3. Then we declared a variable named vpEmail and initialized it with the value of the third element (lstevens@domain.com) of the second element "[1]" of the array named emailList.

You could also retrieve the information using something like:

```
var title = emailList[1][0]  
var email = emailList[1][2]  
alert("The e-mail address for the " + title + " is: " + email)
```

Array Properties

length

The length property returns the number of elements in an array. The format is `arrayName.length`. The length property is particularly useful when using a loop to cycle through an array. One example would be an array used to cycle banners:

```
var bannerImg = new Array();
  bannerImg[0]="image-1.gif";
  bannerImg[1]="image-2.gif";
  bannerImg[2]="image-3.gif";

var newBanner = 0
var totalBan = bannerImg.length

function cycleBan() {
  newBanner++
  if (newBanner == totalBan) {
    newBanner = 0
  }
  document.banner.src=bannerImg[newBanner]
  setTimeout("cycleBan()", 3*1000)
}
window.onload=cycleBan;
```

This portion is then placed in the body where the banner is to be displayed:

```
&lt;img src="image-1.gif" name="banner">
```

Let's take a look and see what happened here: